AXIAL PISTON PUMPS / PRODUCT INFORMATION

P3 Axial Piston Pump

Start with a compact circuit's flow demand and available drive speed. Use the published curves at matching speed and pressure; separate maximum flow and pressure values do not establish one simultaneous operating point.

The P3 delivers more than double the flow of the P1 in the same compact stainless steel package, and adds a boosted inlet option that opens higher shaft speeds and broader system configurations.



WaterQuest P3 axial piston water hydraulic pump, 316 stainless steel

At a glance

Parameter / value	Parameter / value
Displacement 3.0 cc/rev (fixed)	Max Pressure 160 bar (2,321 psi) continuous
Max Flow 6.0 L/min (1.59 US gpm)	Max Input Power 1.75 kW (2.35 hp)
Weight 1.9 kg (4.19 lb)	

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Parameter	Unboosted (PC / PA)	Boosted (PB)
Displacement	3.0 cc/rev (0.183 in³/rev)	3.0 cc/rev (0.183 in³/rev)
Max Continuous Pressure	160 bar (2,321 psi)	160 bar (2,321 psi)
Max Water Flow	5.4 L/min (1.43 US gpm) at 1800 rpm	6.0 L/min (1.59 US gpm) at 2000 rpm
Max RPM	1800 rpm	2000 rpm
Min / Max Inlet Pressure	Positive at all times; max 2.5 bar (36.3 psi)	2.5 bar (36.3 psi) minimum supply
Max Case Drain (T) Pressure	n/a (2-port)	2.5 bar (36.3 psi) maximum
Max Input Power	1.75 kW (2.35 hp)	1.75 kW (2.35 hp)
Weight	1.9 kg (4.19 lb)	1.9 kg (4.19 lb)
Max Fluid Temperature	50°C (consult WQS above 50°C (122 °F))	
Min Fluid Temperature	2°C (35.6 °F) (antifreeze option available)	
Housing Material	316 Stainless Steel	
P Port	1/4" BSPP	1/4" BSPP (x2 pressure ports)
T Port	3/8" BSPP (tank return)	1/4" BSPP (case drain to sump)
Shaft	8.99/8.97 mm (0.354/0.353 in) dia, Key 3x3 mm (0.118x0.118 in)	
Body Dimensions	63 mm (2.48 in) outer / 48 mm (1.89 in) inner, 84 mm (3.31 in) long	
Shaft Section Length	35 mm (15.80 mm (0.622 in) key engagement)	
Mounting	M4 studs supplied, 2 x 4.5 mm (0.177 in) holes, 20 mm (0.787 in) PCD	
Shaft Rotation	PC = CW, PA = CCW	Either direction (ports reverse)
Inlet Filtration	10 µm (394 µin) nominal, 25 µm (984 µin) absolute, beta-10 = 75	
Certifications	Confirm applicability and supplied documentation: ATEX II 2GD ck T6, ISO 9001 FM 87247, F4N	

Configuration and ordering data

Code	Motor	Pressure Range	Flow @ 1500rpm
P3-030-M03	0.25 kW (0.335 hp)	10 to 20 bar (145 to 290 psi)	~4.5 L/min (1.19 US gpm)
P3-030-M04	0.37 kW (0.496 hp)	40 bar (580 psi)	~4.2 L/min (1.11 US gpm)
P3-030-M05	0.55 kW (0.738 hp)	60 bar (870 psi)	~4.1 L/min (1.08 US gpm)
P3-030-M06	0.75 kW (1.01 hp)	80 bar (1,160 psi)	~4.0 L/min (1.06 US gpm)
P3-030-M07	1.1 kW (1.48 hp)	100 to 120 bar (1,450 to 1,740 psi)	~3.9 L/min (1.03 US gpm)
P3-030-M08	1.5 kW (2.01 hp)	140 to 160 bar (2,031 to 2,321 psi)	~3.5 L/min (0.925 US gpm)

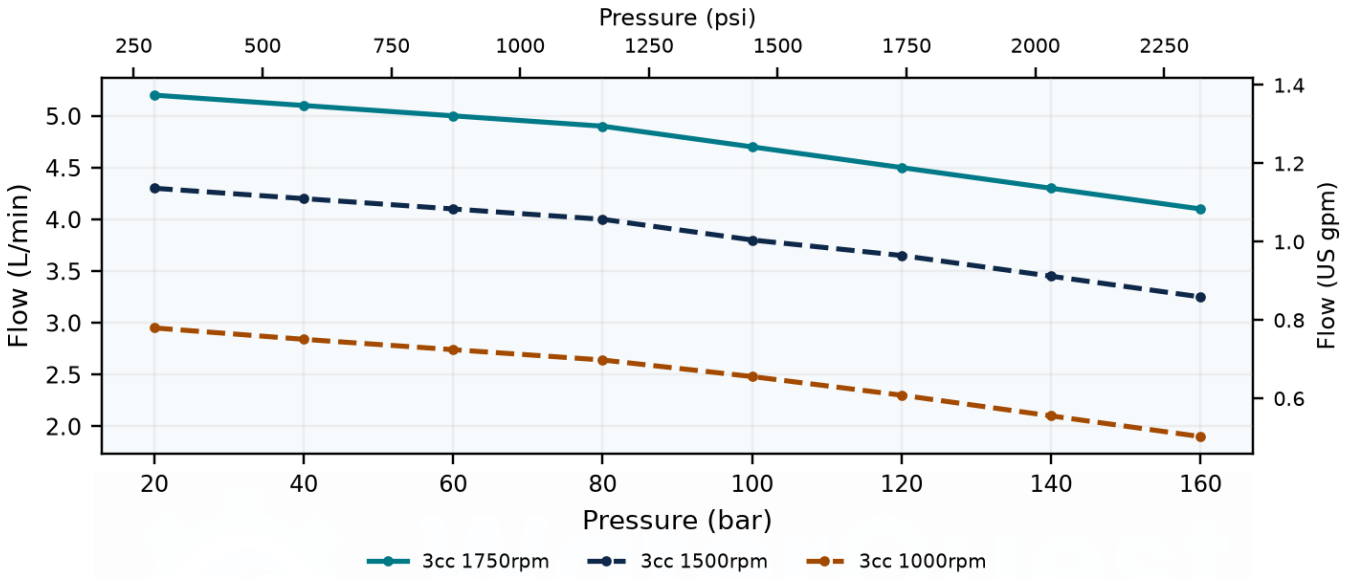
Selection and integration

All performance figures on clean tap water at 15°C (59 °F). The exact operating fluid must be specified on all enquiries and orders. 316 SS MTRs available on request.

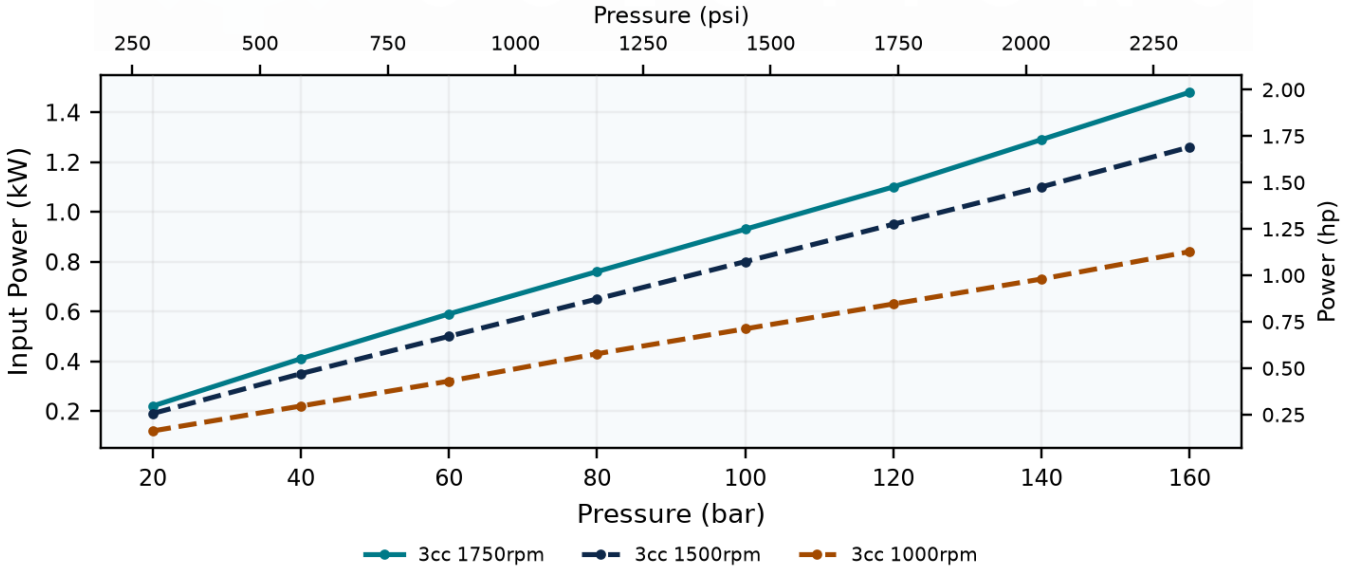
Performance curves

Reference curves reproduced from the product webpage. Curves retain the published data points and speed/displacement labels. Confirm guaranteed performance at the selected duty.

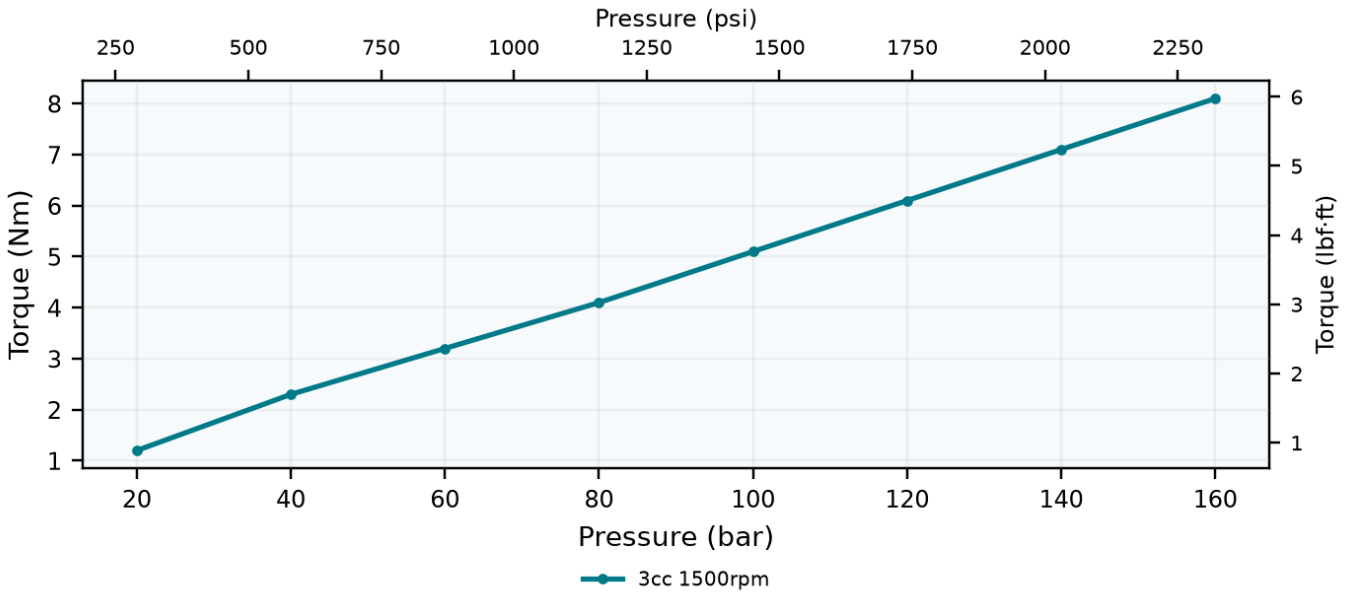
Flow (L/min) vs Pressure (bar)



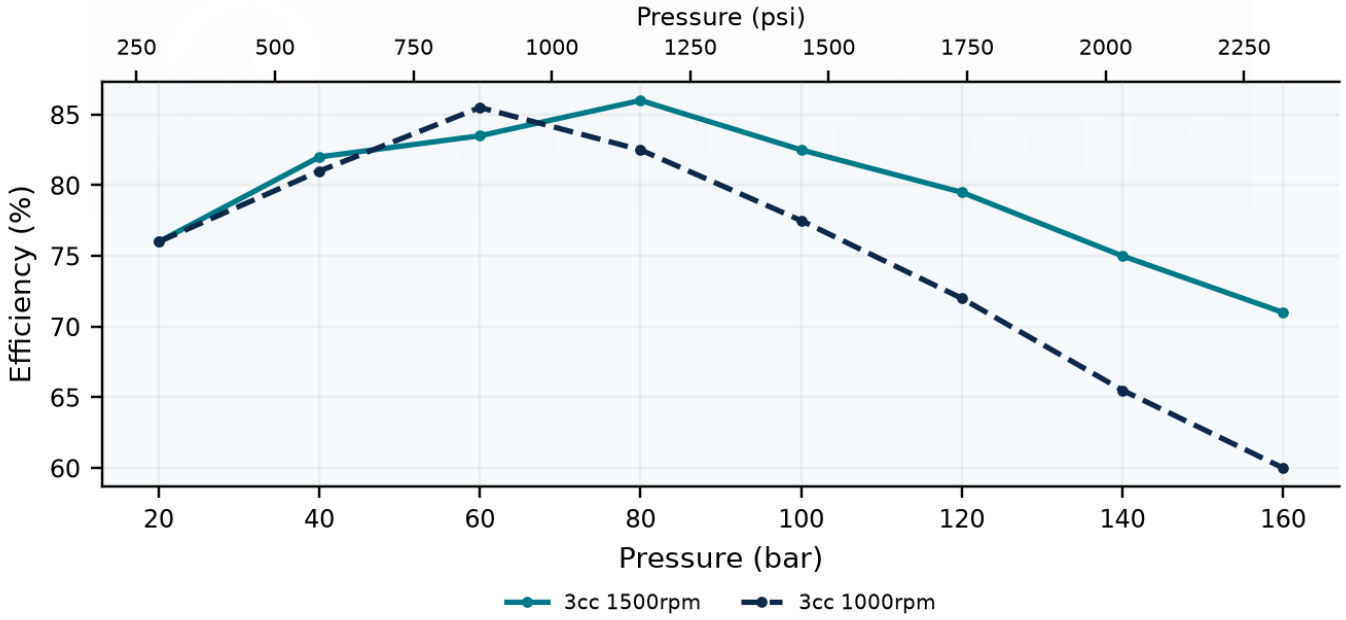
Input Power (kW) vs Pressure (bar)



Torque (Nm) vs Pressure (bar)



Efficiency (%) vs Pressure (bar)

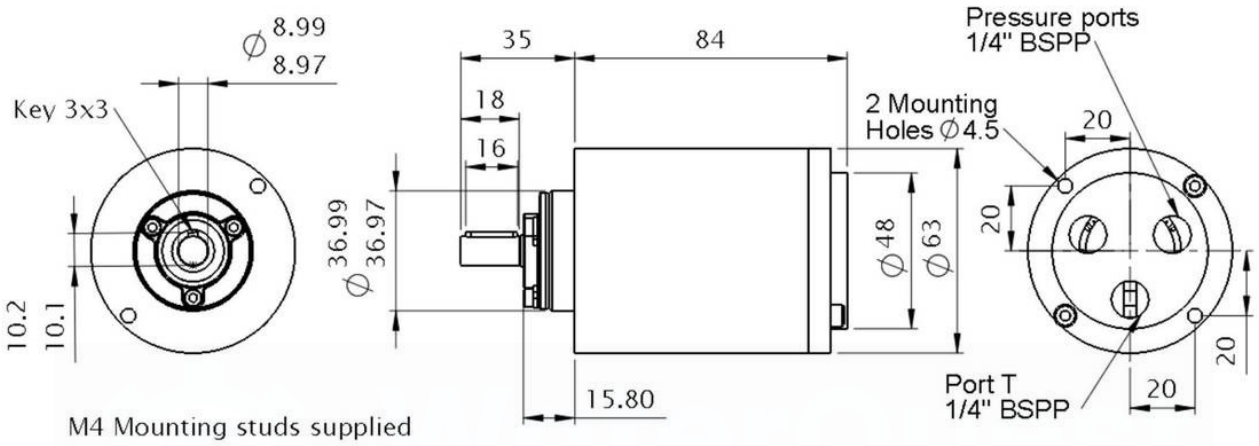


Drawings and reference visuals

Dimensions, ports and curves are reproduced from the webpage visuals. Use the original drawing units and confirm the current approved drawing before fabrication or installation.

PB160-3.0W boosted P3 dimensional drawing, 3-port, 1/4 inch BSPP ports

PB160-3.0W



PC160-3.0W and PA160-3.0W unboosted P3 dimensional drawing, P port 1/4 inch BSPP, T port 3/8 inch BSPP

PC160-3.0W & PA160-3.0W

